Image Sensing and Acquisition

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Illumination and Scene in Image Formation

General Concept

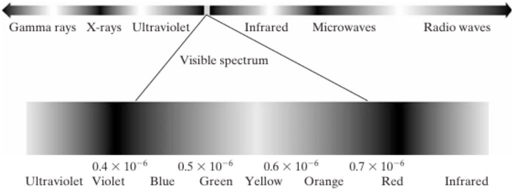
Most images are generated by the interaction of an illumination source with the scene elements via reflection or transmission of energy.

Illumination Sources

- Visible Light
- X-rays
- Infrared
- Radar
- Ultrasound
- Computer-generated patterns

Scene Types

- Natural scenes (buildings, landsca
- Medical structures (e.g., brain)
- Industrial components (molecules,

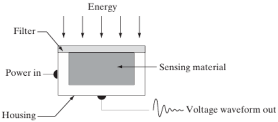

Gamma rays
X-rays
Ultraviolet
Infrared
Microwaves
Radio waves
Visible spectrum
0.4 × 10⁻⁶
0.5 × 10⁻⁶
0.6 × 10⁻⁶
0.7 × 10⁻⁶
Ultraviolet
Violet
Blue
Green
Yellow
Orange
Red
Infrared

Sensor Arrangements for Image Acquisition

Conversion Process

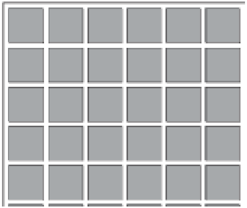
Illumination energy is converted to a voltage signal by sensor materials and digitized into digital form.

Main Sensor Types



(a) Single Sensor

(b) Line Sensor



(c) 2D Array Sensor

Image Acquisition: Single Sensor

How it Works

- Light hits a photodiode sensor.
- Voltage is produced proportional to light intensity.
- A filter may be used to select specific bands (e.g., green light).

Generating 2D Image

- Use relative motion (x and y directions).
- Sensor mounted on screw mechanism or moving bed.
- High precision but slow.

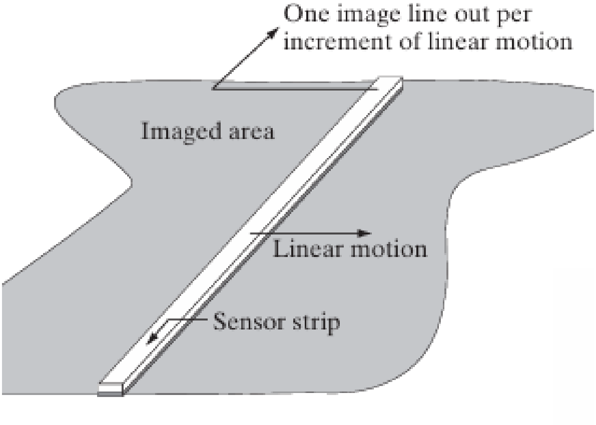
Image Acquisition: Sensor Strip

Description

- 1D strip of sensors in-line.
- Captures one line of the image at a time.
- Motion in perpendicular direction completes 2D image.

Applications

- Flatbed Scanners
- Aerial Mapping from Aircraft
- Multispectral Imaging


One image line out per
increment of linear motion
Imaged area
Linear motion
Sensor strip

Sensor Strip in Circular Geometry

CAT/MRI Imaging

- X-ray or other energy source rotates around the object.
- Opposite sensors detect transmitted energy.
- Data is processed to reconstruct cross-sectional images.

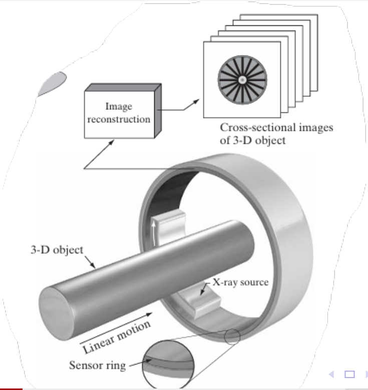


Image Acquisition Using Sensor Arrays

Array Sensors

- 2D matrix of sensor elements.
- Most common: CCD, CMOS.
- Used in digital cameras, telescopes.

Advantages

- Entire image captured at once.
- No mechanical movement required.
- Excellent for low-noise long exposures.

Image Formation Model

Mathematical Model

An image can be represented as a function of illumination and reflectance:

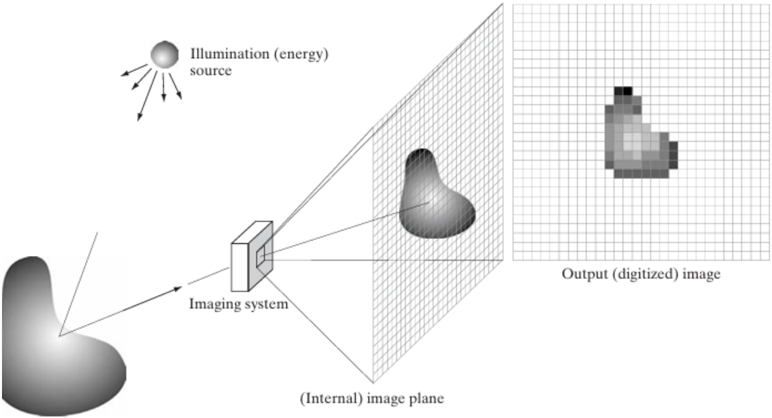
$$f(x, y) = i(x, y) \cdot r(x, y)$$

- $i(x, y)$: Illumination (incident light)
- $r(x, y)$: Reflectance (object surface properties)

Properties

- $0 \leq i(x, y) \leq \infty$
- $0 \leq r(x, y) \leq 1$
- $f(x, y)$ represents the image intensity.

Digital Image Acquisition Pipeline



Steps

1. Energy emitted/reflected from object.
2. Focused onto image plane via optics.
3. Converted to electrical signal by sensor array.
4. Signal digitized to create digital image.